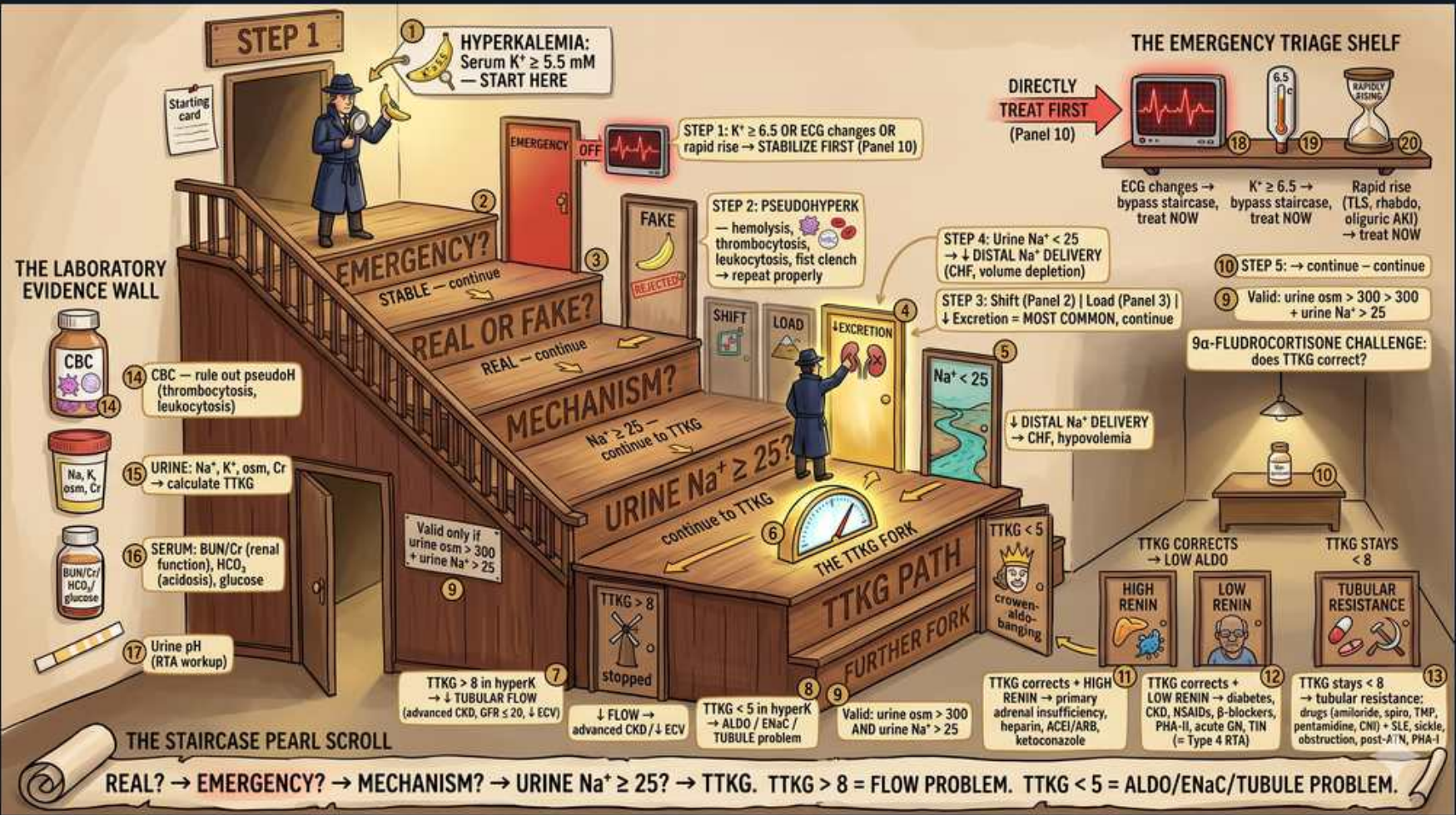


Hyperkalemia — diagnosis (staircase algorithm)



1. K+ >= 5.5 start

WHAT YOU SEE

The detective at the top of the staircase holding a 'STARTING CARD' banana labelled K+ ≥ 5.5 — the first step you climb only once hyperkalemia is confirmed.

WHAT IT ENCODES

Start the workup once serum K+ ≥ 5.5 mM confirms hyperkalemia (mild 5.5–6.0, moderate 6.1–6.9, severe ≥7.0). 98% of total body potassium is intracellular, so even small serum changes reflect large shifts; the staircase then asks: emergency? real? what mechanism?

BOARD TRAP

WRONG answer = treat aggressively off a single borderline value without confirming it is real and not pseudohyperkalemia. Confirm with a repeat sample (and check the ECG) before committing to an etiology — unless the patient already has emergency features.

2. Emergency first

WHAT YOU SEE

The red EMERGENCY? door on the staircase with an ECG-strip and a 'treat first' sign — the alarm door you must go through (stabilize) before continuing the climb.

WHAT IT ENCODES

Identify the emergency immediately: K+ ≥ 6.5 mM, ANY ECG changes, or a rapidly rising level → STABILIZE and TREAT FIRST (membrane stabilization with IV calcium gluconate/chloride, then shift K+ intracellularly with insulin+glucose and albuterol, then remove it with loop diuretics/dialysis). Don't wait for the full diagnostic workup.

BOARD TRAP

WRONG answer = complete the entire diagnostic algorithm (TTKG, fludrocortisone challenge) before treating a patient with K+ 7.2 and peaked T waves. Membrane stabilization and K+ lowering come FIRST; the etiologic workup follows once the patient is safe.

3. Real or fake?

WHAT YOU SEE

The REAL OR FAKE? landing where the detective inspects a banana for a FAKE/REJECTED stamp — the suspicious banana pictures ruling out spurious pseudohyperkalemia.

WHAT IT ENCODES

Step 2 — exclude PSEUDOhyperkalemia (a spuriously high lab value from K+ released in vitro): hemolysis of the sample, prolonged tourniquet time or fist clenching, marked thrombocytosis, or leukocytosis. The patient is asymptomatic with a NORMAL ECG.

BOARD TRAP

WRONG answer = treat a 'high' potassium in a patient with thrombocytosis/leukocytosis and a normal ECG. If pseudohyperkalemia is suspected, redraw properly (a PLASMA sample, no tourniquet/fist clenching, prompt processing) — no treatment is needed. The discriminator: normal ECG + a clotting/cellular cause = fake.

4. Mechanism?

WHAT YOU SEE

The MECHANISM? step where the staircase splits — the fork in the climb cueing the three-bucket sort (shift vs load vs excretion).

WHAT IT ENCODES

Once real, classify the mechanism into one of three buckets: transcellular SHIFT (redistribution out of cells), increased LOAD (intake/release), or decreased renal EXCRETION. This framing directs the next test (history/labs for shift and load; urine studies/TTKG for excretion).

BOARD TRAP

WRONG answer = anchor on one mechanism and stop. Hyperkalemia is often multifactorial (e.g. a CKD patient on an ACEi who develops acidosis). Work through all three buckets rather than attributing it to a single cause.

5. Shift / load

WHAT YOU SEE

The SHIFT (Panel 3) and LOAD (Panel 2) side-doors branching off the stairs — two of the three mechanism buckets drawn as separate doorways.

WHAT IT ENCODES

SHIFT (redistribution): metabolic acidosis (especially mineral/non-anion-gap), insulin deficiency/hyperglycemia, beta-blockers, digoxin toxicity (Na-K-ATPase block), succinylcholine, tumor lysis/rhabdomyolysis, and hyperkalemic periodic paralysis. LOAD: excess oral/IV intake, transfusion of stored blood, GI bleed reabsorption — load alone rarely causes hyperkalemia unless excretion is also impaired.

BOARD TRAP

WRONG answer = blame dietary potassium load in a patient with normal kidneys. Healthy kidneys excrete a large potassium load easily; significant hyperkalemia from load almost always requires concurrent impaired excretion (CKD, low aldosterone, K-sparing drugs).

6. Excretion most common

WHAT YOU SEE

The EXCRETION doorway drawn larger / highlighted as the main path — its prominence marks decreased renal excretion as the most common true mechanism.

WHAT IT ENCODES

DECREASED RENAL EXCRETION is the most common true mechanism of sustained hyperkalemia. It arises from low GFR/low distal flow (advanced CKD, oliguric AKI) or from impaired distal secretion (hypoaldosteronism, aldosterone resistance, ENaC blockade). Urine studies and the TTKG help localize the defect.

BOARD TRAP

WRONG answer = stop at 'renal failure' as the explanation in a patient with only mild CKD. Mild-to-moderate CKD alone rarely causes hyperkalemia until GFR is quite low — look for a superimposed secretory defect (drugs, hypoaldosteronism, type 4 RTA).

7. Urine Na+ >= 25?

WHAT YOU SEE

The URINE Na+ ≥ 25 checkpoint step — the gate that decides whether there is enough distal sodium delivery for the TTKG below to be valid.

WHAT IT ENCODES

Check distal sodium delivery: distal K+ secretion via ENaC requires adequate Na+ reaching the collecting duct. Urine Na+ ≥ 25 mM indicates sufficient distal delivery, making the TTKG interpretable as a measure of aldosterone-driven secretion.

BOARD TRAP

WRONG answer = compute and trust a TTKG when urine Na+ < 25. Low distal Na+ delivery (CHF, hypovolemia, cirrhosis) itself limits K+ secretion and INVALIDATES the TTKG; here, restore perfusion/volume rather than chasing an aldosterone defect.

8. TTKG path

WHAT YOU SEE

The TTKG PATH step with a gauge/dial — the meter pictures measuring aldosterone-driven distal K⁺ secretion.

WHAT IT ENCODES

TTKG (transtubular potassium gradient) = (urine K⁺ / plasma K⁺) ÷ (urine osm / plasma osm). It estimates aldosterone-driven distal K⁺ secretion. Valid ONLY when urine osmolality > serum osmolality AND urine Na⁺ > 25 mM. A normal response to hyperkalemia is a TTKG > 8–10.

BOARD TRAP

WRONG answer = interpret a TTKG without first confirming its validity conditions (urine osm > serum, urine Na⁺ > 25). A 'low' TTKG from low flow or dilute urine will be misread as hypoaldosteronism. (Note: TTKG is a teaching/exam tool; it has fallen out of clinical favor but remains board-relevant.)

9. TTKG > 8 = flow

WHAT YOU SEE

The fork branch labelled TTKG > 8 → FLOW problem (advanced CKD / low GFR) — high gradient drawn as 'secretion works, flow is the issue.'

WHAT IT ENCODES

TTKG > 8 during hyperkalemia means the distal nephron IS secreting K⁺ appropriately (aldosterone and ENaC are working) — so the problem is inadequate distal FLOW/delivery: advanced CKD/ESRD or oliguric AKI. Management centers on flow (diuretics if responsive) or dialysis.

BOARD TRAP

WRONG answer = give fludrocortisone for a high-TTKG hyperkalemia. A TTKG > 8 already shows adequate aldosterone effect — more mineralocorticoid won't help; the issue is low flow/GFR, not hormone deficiency.

10. TTKG < 5 = aldo/tubule

WHAT YOU SEE

The fork branch labelled TTKG < 5 → ALDO / ENaC / tubule problem — low gradient drawn as failed distal secretion.

WHAT IT ENCODES

TTKG < 5 in the face of hyperkalemia = inappropriately LOW distal K⁺ secretion → an aldosterone problem (deficiency or resistance), an ENaC defect, or a tubular/tubulointerstitial disorder. The next step is the fludrocortisone challenge to separate hormone deficiency from tubular resistance.

BOARD TRAP

WRONG answer = call any low TTKG 'hypoaldosteronism.' A low TTKG could be aldosterone DEFICIENCY (corrects with fludrocortisone) OR tubular RESISTANCE/ENaC blockade (does not correct). The fludrocortisone test is what distinguishes them.

11. Fludrocortisone test

WHAT YOU SEE

The 9-alpha-FLUDROCORTISONE CHALLENGE bench — giving exogenous mineralocorticoid to see whether the TTKG corrects.

WHAT IT ENCODES

9-alpha-fludrocortisone challenge: give exogenous mineralocorticoid and re-measure TTKG. If secretion (TTKG) corrects/rises, the collecting duct is responsive and the defect was aldosterone deficiency. If TTKG stays low, the tubule is resistant to mineralocorticoid.

BOARD TRAP

WRONG answer = use the fludrocortisone challenge before confirming the TTKG was valid (urine Na⁺ > 25, concentrated urine). Testing on an invalid TTKG gives uninterpretable results.

12. Corrects -> low aldo

WHAT YOU SEE

The 'TTKG CORRECTS → LOW ALDO' branch splitting into HIGH RENIN vs LOW RENIN — correction pictures a responsive duct, so the defect was aldosterone deficiency.

WHAT IT ENCODES

If TTKG CORRECTS with fludrocortisone → aldosterone DEFICIENCY. Then check renin: HIGH renin/low aldosterone = primary adrenal failure (Addison disease, isolated hypoaldosteronism); LOW renin/low aldosterone = hyporeninemic hypoaldosteronism (type 4 RTA), classically diabetic nephropathy, NSAIDs, or calcineurin inhibitors.

BOARD TRAP

WRONG answer = treat all aldosterone deficiency the same. The renin level branches it: low-renin (type 4 RTA, diabetes/NSAIDs) is managed differently (fludrocortisone, loop diuretic, low-K diet) than primary adrenal insufficiency (which also needs glucocorticoid replacement).

13. No correction -> tubule

WHAT YOU SEE

The 'TTKG STAYS < 8' branch labelled TUBULAR RESISTANCE — no correction pictures a duct deaf to aldosterone (PHA, ENaC-blocking drugs, tubulointerstitial disease).

WHAT IT ENCODES

If TTKG does NOT correct with fludrocortisone → tubular RESISTANCE to aldosterone: pseudohypoaldosteronism (PHA type 1/2), drugs blocking ENaC (amiloride, triamterene, trimethoprim, pentamidine), or tubulointerstitial disease (obstruction, sickle cell, lupus, amyloid).

BOARD TRAP

WRONG answer = keep escalating mineralocorticoid when the tubule won't respond. In aldosterone resistance, more fludrocortisone is futile — remove the offending ENaC-blocking drug and treat the underlying tubulointerstitial process; use a loop/thiazide diuretic to enhance K⁺ loss.

14. Lab evidence wall

WHAT YOU SEE

The LABORATORY EVIDENCE WALL hung with bottles — CBC, urine lytes, BUN/Cr, HCO₃, glucose — the core panel that sorts pseudo vs shift vs load vs excretion.

WHAT IT ENCODES

Core lab panel for the workup: CBC (thrombocytosis/leukocytosis → pseudohyperkalemia), urine Na⁺/K⁺/osmolality (distal delivery and TTKG), BUN/creatinine (GFR/renal cause), bicarbonate (metabolic acidosis driving shift or type 4 RTA), and glucose (hyperglycemia/insulin-deficiency shift). Plus an ECG in every case.

BOARD TRAP

WRONG answer = order potassium-lowering treatment without an ECG and without these confirmatory labs. The ECG dictates urgency (and may be the only sign of life-threatening hyperkalemia), while the labs distinguish pseudo, shift, load, and excretion mechanisms.